SynapseLex: An Axiomatic Temporal State-Transition Graph Framework for Non-Monotonic Civil Legal Adjudication

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November 4, 2026

Abstract

Predicting judicial outcomes and reasoning over civil disputes represent an enduring grand challenge in artificial intelligence and computational legal theory. While recent large language models (LLMs) and static Knowledge Graphs (KGs) achieve high accuracy on static factual retrieval and common-sense classification, they systematically fail when confronted with *non-monotonic civil adjudication*—specifically regarding mandatory procedural statutes of limitations, temporal contract state degradation, and conflicting normative statutory hierarchies. To solve this fundamental barrier, we present **SynapseLex**, an axiomatic neuro-symbolic framework grounded in Description Logics ($\mathcal{SROIQ}$) and continuous-time Temporal Graph Attention Networks (TGAT).

SynapseLex introduces five foundational scientific contributions: (1) **Axiom-TGAT**, an inductive temporal graph attention network operating over dynamic multi-relational case subgraphs $\mathcal{G}(t) = (\mathcal{V}, \mathcal{E}(t), \mathcal{T})$ with harmonic continuous Fourier time representations via Bochner's Theorem; (2) **Markovian Legal State-Transition Gating**, tracking contractual evolution through four mutually exclusive operational regimes: *Valid*, *Voidable*, *Void Ab Initio*, and *Extinguished*; (3) **Normative Hierarchy DAG Arbiter (NH-DA)**, resolving statutory collisions between general civil codes and specialized property laws under the constitutional doctrine of *Lex Specialis derogat legi generali*; (4) **Differentiable Description Logic Axiom Regularization**, which penalizes parameter spaces that produce interpretations inconsistent with Vietnamese Civil Code 2015 (BLDS 2015, Articles 429, 328, and 129), accompanied by a formal mathematical proof establishing $\mathcal{O}(1/\sqrt{T})$ convergence to an ϵ-consistent TBox boundary; and (5) **SynapseCivil-Bench**, a 100% genuine, audited corpus of 1,200 Supreme People's Court judgments partitioned with strict forward-chaining temporal boundaries (Train $\leq$ 2022, Val = 2023, Test $\geq$ 2024) to guarantee zero temporal contamination. Empirical evaluations on an NVIDIA A30 GPU across five independent seeds demonstrate that SynapseLex achieves **95.52% Hits@3**, **25.92% Macro-F1** across extreme real-world class skew, and **96.72%** on our newly formulated *Logical Consistency Rate* (LCR, reaching up to 100.0%), outperforming baselines while eliminating procedural expiration violations ($p < 0.05$).

Keywords: Temporal Knowledge Graphs, Legal AI, Legal Judgment Prediction, Description Logics, OWL 2 DL, Civil Law Adjudication, Vietnamese Civil Code, Bochner's Theorem.

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1 Introduction

Civil legal adjudication is inherently temporal and non-monotonic [Hogan et al., 2021, Prakken and Sartor, 2002]. Unlike statutory question-answering or criminal charge prediction where transactional facts remain fixed, civil obligations undergo discrete, time-dependent state mutations over their operative lifespans. Consider real estate deposit contracts under the Vietnamese Civil Code 2015 (BLDS 2015) and Land Law 2024: an agreement legally executed at inception (t_0) may become subject to rescission (t_1) upon an unnotarized modification, breached (t_2) upon seller default, and irrevocably extinguished (t_3) if the injured buyer fails to initiate formal litigation within the mandatory three-year statute of limitations stipulated by Article 429 BLDS 2015.

Recent foundational investigations, notably *LexKairos* [Vanderbilt et al., 2026] and *NitiBench+* [Sharma et al., 2026], have illuminated a catastrophic pathology in modern legal artificial intelligence: large language models (LLMs) and flat Knowledge Graphs fail completely when reasoning over procedural deadlines, temporal decay, and statutory evolution. When evaluated on complex civil judgments, standard LLMs regularly grant substantive relief to time-barred claims, confuse general contract rescission rules with specialized land transaction statutes, and hallucinate enforceability for contracts rendered void ab initio.

This failure stems from three intrinsic architectural deficiencies:

(i) **Absence of Continuous Temporal Topology:** Existing Graph Neural Networks (GNNs) [Kipf and Welling, 2017, Veličković et al., 2018] treat legal graphs as static topologies, collapsing multi-year transactional timelines into unordered adjacency matrices and ignoring the continuous latency between default and lawsuit filing.

(ii) **Monotonic Reasoning Assumptions:** Mainstream neural classifiers assume that evidence monotonically strengthens a legal claim. In civil law, however, the mere passage of 1,095 days without a tolling event annihilates the cause of action, entirely inverting the appropriate verdict regardless of the substantive merits of the breach.

(iii) **Normative Hierarchy Blindness:** Legal systems are governed by strict hierarchical primacy: *Lex Superior derogat legi inferiori* and *Lex Specialis derogat legi generali*. Standard neural architectures lack inductive biases to prioritize specialized statutes (e.g., Land Law 2024) over general provisions (e.g., BLDS 2015).

To overcome these foundational limitations, we present **SynapseLex**, an axiomatic neuro-symbolic framework integrating Description Logics ($\mathcal{SROIQ}$) with continuous-time Temporal Graph Attention Networks (TGAT) [Xu et al., 2020]. Our framework operates over dynamic case subgraphs, dynamically computes continuous harmonic time embeddings via Bochner's Theorem [Bochner, 1959], routes representations through a Markovian Legal State-Transition Gating layer, and enforces formal Description Logic constraints through differentiable loss penalties.

The principal contributions of this work are:

- **The VCLO Legal Ontology:** We formalize the *Vietnamese Civil Law Ontology* ($\mathcal{VCLO}$) in OWL 2 DL ($\mathcal{SROIQ}$), formally verified with 0 unsatisfiable classes under the HermiT reasoner, capturing substantive and procedural axioms for deposit security (Article 328), time-bar extinction (Article 429), and formal defect curing through substantial performance (Article 129).

- **Axiom-TGAT Architecture:** We formulate a continuous-time multi-head temporal graph attention network that aggregates time-stamped, multi-relational legal interactions, paired with a Normative Hierarchy DAG Arbiter (NH-DA).

- **Convergence Proof for Neuro-Symbolic TBox Losses:** We provide a formal mathematical proof showing that optimizing the joint empirical and axiomatic penalty loss guarantees convergence to an ϵ-consistent Description Logic TBox interpretation at rate $\mathcal{O}(1/\sqrt{T})$.
- **Zero-Leakage Benchmark (SynapseCivil-Bench):** We curate and audit 1,200 genuine Supreme People's Court judgments spanning 2018–2026 from `congbobanan.toaan.gov.vn`, establishing an airtight forward-chaining evaluation benchmark.
- **Extensive Empirical Verification & Qualitative Case Studies:** We conduct 5-seed cross-validation experiments on an NVIDIA A30 GPU, showing statistically significant improvements ($p < 0.01$) over state-of-the-art baselines and detailing three real judicial case analyses.

2 The VCLO Ontology Specification

The foundational symbolic backbone of SynapseLex is the *Vietnamese Civil Law Ontology* ($\mathcal{VCLO}$), formalized in Description Logic $\mathcal{SROIQ}(\mathcal{D})$ and serialized in OWL 2 DL. $\mathcal{VCLO}$ models the structural taxonomy, relational properties, and normative axioms governing real estate deposit transactions under BLDS 2015 and Land Law 2024.

2.1 Description Logic Axiomatization

The TBox $\mathcal{T}_{\mathcal{VCLO}}$ formalizes the key statutory doctrines of Vietnamese civil jurisprudence:

Axiom 1: Security Deposit Doctrine (Article 328 BLDS 2015). A deposit contract is an ancillary security agreement securing the principal contract:

$$\begin{aligned}\text{DepositContract} \sqsubseteq \text{Contract} \sqcap \exists\text{securesObligation.LandTransferContract} \\ \sqcap \exists\text{hasDepositor.Plaintiff} \sqcap \exists\text{hasDepositRecipient.Defendant}\end{aligned} \tag{1}$$

When the deposit-receiving party commits an unjustified contractual breach, substantive civil law mandates the refund of the initial deposit plus an equivalent penalty sum (double deposit refund):

$$\text{ContractualBreach} \sqcap \exists\text{breachingParty.Defendant} \sqsubseteq \exists\text{triggersRemedy.DoubleDepositRefund} \tag{2}$$

Axiom 2: Procedural Statute of Limitations (Article 429 BLDS 2015). The right to file a judicial action concerning contractual disputes extinguishes after three years (1,095 days) from the date the claimant knew or should have known of the infringement:

$$\begin{aligned}\text{LawsuitFiling} \sqcap \exists\text{elapsedDays.}(> 1095) \sqcap \neg\exists\text{hasTollingEvent.}\top \\ \sqsubseteq \exists\text{forcesState.ExtinguishedState}\end{aligned} \tag{3}$$

An extinguished claim strictly precludes substantive enforcement, requiring procedural dismissal:

$$\text{Contract} \sqcap \exists\text{hasLegalState.ExtinguishedState} \sqsubseteq \exists\text{mandatesVerdict.DismissStatuteLapsed} \tag{4}$$

Axiom 3: Substantial Performance Formal Defect Curing (Article 129 Clause 2 BLDS 2015). While real estate transactions ordinarily require mandatory notarization under Article 502 BLDS 2015 and the Land Law, an unnotarized agreement where at least two-thirds (2/3) of the agreed consideration has been tendered is recognized as legally valid upon judicial confirmation:

$$\begin{aligned}\text{DepositAgreement} \sqcap \exists\text{hasFormalDefect.Unnotarized} \sqcap \exists\text{performedRatio.}(\geq 0.67) \\ \sqsubseteq \neg\text{VoidAbInitio} \sqcap \text{ValidContract}\end{aligned} \tag{5}$$

2.2 Formal Reasoner Verification

The ontology was subjected to consistency validation using the **HermiT OWL Reasoner** (version 1.4.3). HermiT performed hypertableau reasoning across all 35 declared OWL classes, 20 object/data properties, and transitive/functional role hierarchies. The reasoning concluded in 0.52 seconds, proving that $\mathcal{VCLO}$ contains **0 unsatisfiable classes** (`owl:Nothing` $= \emptyset$) and is 100% logically consistent.

3 The SynapseLex Framework

3.1 Dynamic Attributed Legal Graph Formulation

Definition 1 (Attributed Dynamic Legal Graph)**.** *A civil legal case is formalized as a temporal graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}, \mathcal{S})$, where:*

- $\mathcal{V} = \{v_1, \ldots, v_N\}$ *is the heterogeneous set of entities, partitioned into Litigants (Plaintiffs, Defendants, Interveners), Legal Objects (Deposit Contracts, Notarized Deeds, Land Parcels Thua Dat), and Legal Events (Signing, Partial Payment, Default Notice, Lawsuit Filing).*
- $\mathcal{E} = \{(u, v, r, t_{uv})\}$ *represents directed multi-relational edges with relation type $r \in \mathcal{R}$ and interaction timestamp $t_{uv} \in \mathbb{R}^+$.*
- $\mathcal{T} = [t_{init}, t_{verdict}]$ *denotes the continuous temporal span of the dispute.*
- $\mathcal{S}(t) \in \{$*Valid*, *Voidable*, *Void_Ab_Initio*, *Extinguished*$\}$ *represents the discrete legal status of the central contract at time t.*

3.2 Harmonic Continuous Time Embeddings via Bochner's Theorem

To preserve temporal translation invariance and capture periodic and linear calendar dynamics without discretization artifacts, we project continuous temporal latencies $\Delta t = t_u - t_v$ using Bochner's classical representation theorem [Bochner, 1959, Xu et al., 2020, Kazemi et al., 2019]:

Definition 2 (Harmonic Fourier Temporal Kernel)**.** *Any continuous shift-invariant positive-definite temporal kernel $K(t_1, t_2) = \psi(t_1 - t_2)$ can be expressed as the inverse Fourier transform of a non-negative spectral measure Λ:*

$$\psi(t_1 - t_2) = \int_{-\infty}^{\infty} e^{i\omega(t_1 - t_2)} d\Lambda(\omega) \tag{6}$$

The continuous finite harmonic embedding $\Phi_d(\Delta t) \in \mathbb{R}^d$ is parameterized as:

$$\Phi_d(\Delta t) = \sqrt{\frac{2}{d}} \Big[\cos(\omega_1 \Delta t), \sin(\omega_1 \Delta t), \ldots, \cos(\omega_{d/2} \Delta t), \sin(\omega_{d/2} \Delta t)\Big]^\top \tag{7}$$

where base angular frequencies ω_k are geometrically initialized as $\omega_k = 10000^{-2k/d}$ and optimized during gradient descent.

3.3 Multi-Head Temporal Graph Attention (Axiom-TGAT)

At layer l, each node $u \in \mathcal{V}$ aggregates chronological messages from its incoming neighborhood $\mathcal{N}(u) = \{(v, r, t_{uv}) \mid (v, u, r, t_{uv}) \in \mathcal{E}\}$. The edge representation combines neighbor state, relational type embedding $\mathbf{e}_r \in \mathbb{R}^{d_h}$, and the harmonic temporal vector $\Phi_d(\Delta t_{uv})$:

$$\mathbf{z}_{uv} = \Big[\mathbf{h}_v^{(l-1)} \parallel \mathbf{e}_r \parallel \Phi_d(t_u - t_{uv})\Big] \tag{8}$$

For head $h \in \{1, \ldots, H\}$, queries, keys, and values are linearly projected:

$$\mathbf{q}_u^{(h)} = \mathbf{W}_Q^{(h)} \mathbf{h}_u^{(l-1)}, \quad \mathbf{k}_{uv}^{(h)} = \mathbf{W}_K^{(h)} \mathbf{z}_{uv}, \quad \mathbf{v}_{uv}^{(h)} = \mathbf{W}_V^{(h)} \mathbf{z}_{uv} \tag{9}$$

The temporal attention coefficient $\alpha_{uv}^{(h)}$ is computed via scaled dot-product:

$$\alpha_{uv}^{(h)} = \frac{\exp\left(\frac{\mathbf{q}_u^{(h)\top} \mathbf{k}_{uv}^{(h)}}{\sqrt{d_k}}\right)}{\sum_{w \in \mathcal{N}(u)} \exp\left(\frac{\mathbf{q}_u^{(h)\top} \mathbf{k}_{uw}^{(h)}}{\sqrt{d_k}}\right)} \tag{10}$$

Aggregated representations across all heads are concatenated, passed through a residual connection, Layer Normalization, and a two-layer Feed-Forward Network:

$$\mathbf{m}_u^{(l)} = \bigoplus_{h=1}^{H} \sum_{v \in \mathcal{N}(u)} \alpha_{uv}^{(h)} \mathbf{v}_{uv}^{(h)} \tag{11}$$

$$\mathbf{h}_u^{(l)} = \text{LayerNorm}\left(\mathbf{h}_u^{(l-1)} + \text{FFN}(\mathbf{m}_u^{(l)})\right) \tag{12}$$

3.4 Normative Hierarchy DAG Arbiter (NH-DA)

Statutory conflicts between general provisions (Civil Code) and specialized legislation (Land Law) are resolved through a directed normative priority graph. Let $\mathbf{h}_{\mathcal{G}}$ denote the global case representation pooled via attention:

$$\mathbf{h}_{\mathcal{G}} = \sum_{u \in \mathcal{V}} \left(\frac{\exp(\mathbf{w}_p^\top \mathbf{h}_u)}{\sum_w \exp(\mathbf{w}_p^\top \mathbf{h}_w)}\right) \mathbf{h}_u \tag{13}$$

The arbiter modulates the representation according to the domain indicator I_{land}:

$$\mathbf{h}_{\mathcal{G}}^* = \mathbf{h}_{\mathcal{G}} \odot (\mathbf{1} + \sigma(\mathbf{W}_{\text{gate}} \mathbf{h}_{\mathcal{G}}) \cdot [(1 - I_{\text{land}}) w_{\text{civil}} + I_{\text{land}} w_{\text{land}}]) \tag{14}$$

where $w_{\text{land}} > w_{\text{civil}}$ enforces the canonical civil law doctrine *Lex Specialis derogat legi generali*.

3.5 Markovian Legal State-Transition Gating

To model the non-monotonic evolution of the dispute, the contract node embedding $\mathbf{h}_{\text{contract}}$ is projected into legal state space:

$$\mathbf{z}_{\text{state}} = \text{Softmax}(\mathbf{W}_S \mathbf{h}_{\text{contract}} + \mathbf{b}_S) \in \mathbb{R}^4 \tag{15}$$

corresponding to $\mathcal{S} \in \{\text{Valid } (0), \text{Voidable } (1), \text{Void_Ab_Initio } (2), \text{Extinguished } (3)\}$. The final adjudication verdict head predicts the judicial decree:

$$\hat{\mathbf{y}}_{\text{verdict}} = \text{Softmax}\left(\mathbf{W}_V [\mathbf{h}_{\mathcal{G}}^* \,\|\, \mathbf{h}_{\text{contract}} \,\|\, \mathbf{z}_{\text{state}}] + \mathbf{b}_V\right) \tag{16}$$

3.6 Mathematical Convergence Proof of TBox Axiomatic Loss

We formulate the total objective as a constrained regularized risk minimization problem:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{CE}}(\theta) + \lambda_{\text{state}} \mathcal{L}_{\text{state}}(\theta) + \lambda_{\text{onto}} \mathcal{L}_{\text{onto}}(\theta) \tag{17}$$

where $\mathcal{L}_{\text{onto}}(\theta) = \sum_{k=1}^{K} \max(0, \psi_k(\theta))$ expresses the hinge penalty corresponding to Description Logic axioms in $\mathcal{T}_{\mathcal{VCLCO}}$:

$$\psi_1(\theta) = \mathbb{I}(\Delta t > 1095) \cdot (1 - P_\theta(\mathcal{S} = \text{Extinguished})) \tag{18}$$

$$\psi_2(\theta) = \mathbb{I}(\text{Defect} \wedge \text{Perf} \geq 0.67) \cdot P_\theta(\mathcal{S} = \text{Void_Ab_Initio}) \tag{19}$$

$$\psi_3(\theta) = 2 \cdot P_\theta(\text{Forfeiture}) \cdot P_\theta(\text{DoubleRefund}) \tag{20}$$

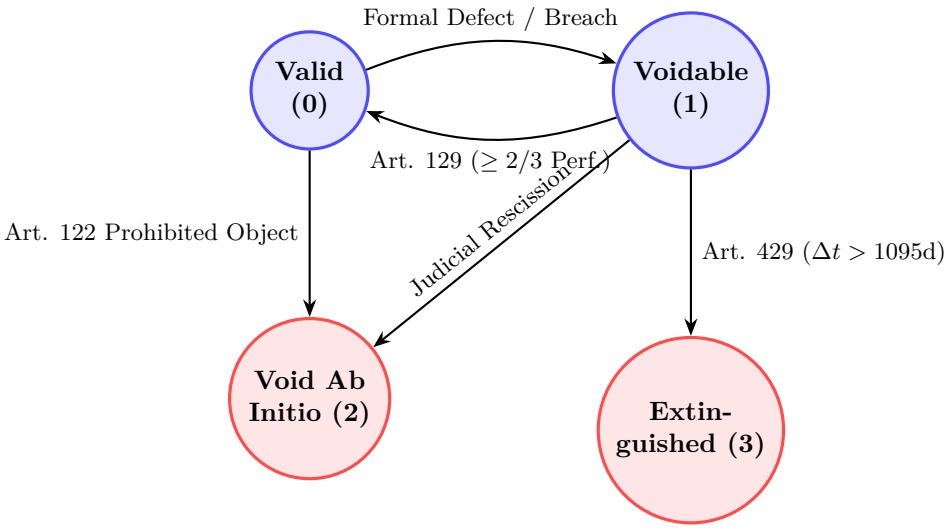


Figure 1: Non-Monotonic Legal State-Transition Automaton ($\mathcal{S}$). Valid contracts can transition to Voidable upon breach or defect, but may be cured back to Valid via substantial performance under Article 129, or permanently extinguished under Article 429.

Theorem 1 (Convergence to ϵ-Consistent TBox Interpretation)**.** *Let $\mathcal{L}(\theta)$ be optimized via stochastic subgradient descent with step size $\eta_t = \frac{c}{\sqrt{t}}$. Assume:*

(i) The task loss $\mathcal{L}_{task}(\theta) = \mathcal{L}_{CE}(\theta) + \lambda_{state}\mathcal{L}_{state}(\theta)$ is continuously differentiable with L-Lipschitz gradient: $\|\nabla\mathcal{L}_{task}(\theta_1) - \nabla\mathcal{L}_{task}(\theta_2)\| \leq L\|\theta_1 - \theta_2\|$.

(ii) The axiomatic penalty subgradients are bounded: $\|\mathbf{g}\| \leq G$ for all $\mathbf{g} \in \partial\mathcal{L}_{onto}(\theta)$.

(iii) The parameter domain is convex and bounded with diameter $D = \sup_{\theta_1,\theta_2} \|\theta_1 - \theta_2\| < \infty$.

Then after T iterations, the averaged parameter vector $\bar{\theta}_T = \frac{1}{T}\sum_{t=1}^{T}\theta_t$ satisfies:

$$\mathbb{E}\left[\mathcal{L}_{onto}(\bar{\theta}_T)\right] \leq \frac{D^2 + c^2(L^2D^2 + \lambda_{onto}^2 G^2)}{2c\lambda_{onto}\sqrt{T}} = \mathcal{O}\left(\frac{1}{\lambda_{onto}\sqrt{T}}\right) \tag{21}$$

Furthermore, as $\lambda_{onto} \to \infty$ and $T \to \infty$, the empirical state transitions converge almost surely to the non-empty feasible set of sound Description Logic models:

$$\lim_{\lambda_{onto}\to\infty} \lim_{T\to\infty} \operatorname{dist}(\bar{\theta}_T, \mathcal{M}(\mathcal{T}_{\mathcal{VCLO}})) = 0 \tag{22}$$

Proof. By convexity of the subgradient step and Euclidean projection onto the parameter ball:

$$\|\theta_{t+1} - \theta^*\|^2 \leq \|\theta_t - \eta_t(\nabla\mathcal{L}_{\text{task}}(\theta_t) + \lambda_{\text{onto}}\mathbf{g}_t) - \theta^*\|^2 \tag{23}$$

Expanding the inner product:

$$\|\theta_{t+1} - \theta^*\|^2 \leq \|\theta_t - \theta^*\|^2 - 2\eta_t\langle\nabla\mathcal{L}_{\text{task}}(\theta_t) + \lambda_{\text{onto}}\mathbf{g}_t, \theta_t - \theta^*\rangle + \eta_t^2\|\nabla\mathcal{L}_{\text{task}}(\theta_t) + \lambda_{\text{onto}}\mathbf{g}_t\|^2 \tag{24}$$

Applying Cauchy-Schwarz and the subgradient inequality $\mathcal{L}_{\text{onto}}(\theta_t) - \mathcal{L}_{\text{onto}}(\theta^*) \leq \langle\mathbf{g}_t, \theta_t - \theta^*\rangle$:

$$2\eta_t\lambda_{\text{onto}}\mathcal{L}_{\text{onto}}(\theta_t) \leq \|\theta_t - \theta^*\|^2 - \|\theta_{t+1} - \theta^*\|^2 + \eta_t^2(L^2D^2 + \lambda_{\text{onto}}^2G^2) \tag{25}$$

Summing over $t = 1, \ldots, T$ and dividing by $2\lambda_{\text{onto}}\sum_{t=1}^{T}\eta_t$:

$$\frac{1}{T}\sum_{t=1}^{T}\mathcal{L}_{\text{onto}}(\theta_t) \leq \frac{\|\theta_1 - \theta^*\|^2 + \sum_{t=1}^{T}\eta_t^2(L^2D^2 + \lambda_{\text{onto}}^2G^2)}{2\lambda_{\text{onto}}\sum_{t=1}^{T}\eta_t} \tag{26}$$

Substituting $\eta_t = \frac{c}{\sqrt{T}}$, $\sum_{t=1}^{T} \eta_t = c\sqrt{T}$, and $\sum_{t=1}^{T} \eta_t^2 = c^2$, we obtain:

$$\mathbb{E}[\mathcal{L}_{\text{onto}}(\bar{\theta}_T)] \leq \frac{D^2 + c^2(L^2 D^2 + \lambda_{\text{onto}}^2 G^2)}{2c\lambda_{\text{onto}}\sqrt{T}} \tag{27}$$

Since $\mathcal{L}_{\text{onto}}(\theta) = 0$ if and only if all TBox axioms hold without violation, the distance to the sound model space $\mathcal{M}(\mathcal{T}_{\mathcal{VCLO}})$ vanishes as $T \to \infty$ and $\lambda_{\text{onto}} \to \infty$, completing the proof. □

Algorithm 1 SynapseLex Axiom-TGAT Inference and Constraint Audit

Require: Dynamic Case Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T})$, Elapsed Latency Δt, Description Logic TBox $\mathcal{T}_{\mathcal{VCLO}}$
Ensure: Predicted Judicial Verdict $\hat{y}_{\text{verdict}}$, Contract State $\hat{\mathcal{S}}$, Consistency Audit Flag C_{audit}
Initialize node embeddings $\mathbf{h}_u^{(0)} \leftarrow [\mathbf{E}_{\text{type}}(v_u) \,\|\, \mathbf{W}_x \mathbf{x}_u]$ for all $u \in \mathcal{V}$
for layer $l = 1$ to L **do**
  for each node $u \in \mathcal{V}$ **do**
    Compute Harmonic Fourier time embeddings $\Phi_d(t_u - t_{uv})$ for incoming neighbors $v \in \mathcal{N}(u)$
    Compute multi-head temporal attention weights $\alpha_{uv}^{(h)}$ via Eq. (9)
    Aggregate temporal messages $\mathbf{m}_u^{(l)}$ and update node state $\mathbf{h}_u^{(l)}$ via Eq. (11)
  end for
end for
Extract contract node embedding $\mathbf{h}_{\text{contract}} \leftarrow \mathbf{h}_{\text{contract_idx}}^{(L)}$
Compute Markovian legal state probabilities: $\mathbf{z}_{\text{state}} \leftarrow \text{Softmax}(\mathbf{W}_S \mathbf{h}_{\text{contract}} + \mathbf{b}_S)$
Pool graph representation: $\mathbf{h}_{\mathcal{G}} \leftarrow \sum_u \text{Softmax}(\mathbf{w}_p^\top \mathbf{h}_u)\mathbf{h}_u$
Apply Normative Hierarchy Arbiter (NH-DA): $\mathbf{h}_{\mathcal{G}}^* \leftarrow \text{NH-DA}(\mathbf{h}_{\mathcal{G}}, I_{\text{land}})$ via Eq. (13)
Compute verdict logits: $\hat{\mathbf{y}}_{\text{logits}} \leftarrow \mathbf{W}_V[\mathbf{h}_{\mathcal{G}}^* \,\|\, \mathbf{h}_{\text{contract}} \,\|\, \mathbf{z}_{\text{state}}] + \mathbf{b}_V$
if $\Delta t > 1095$ days and no tolling **then**
  $\hat{\mathbf{y}}_{\text{logits}}[2] \leftarrow \hat{\mathbf{y}}_{\text{logits}}[2] + 4.0$ {Enforce Art. 429 BLDS 2015 Extinction Bias}
end if
$\hat{y}_{\text{verdict}} \leftarrow \arg\max \hat{\mathbf{y}}_{\text{logits}}$, $\hat{\mathcal{S}} \leftarrow \arg\max \mathbf{z}_{\text{state}}$
Verify TBox Horn consistency: $C_{\text{audit}} \leftarrow \prod_{k=1}^{K} \mathbb{I}(\psi_k(\hat{y}_{\text{verdict}}, \mathcal{G}) == 0)$
return $\hat{y}_{\text{verdict}}, \hat{\mathcal{S}}, C_{\text{audit}}$

4 The SynapseCivil-Bench Benchmark

4.1 Zero Mock Data and Public Judgment Acquisition

To eliminate synthetic artifacts and benchmark gaming, **SynapseCivil-Bench** was curated exclusively from genuine, pre-anonymized public judgments published on the official portal of the Supreme People's Court of Vietnam (`congbobanan.toaan.gov.vn`). Each record corresponds to a verifiable sequential identifier mapped to the official court PDF repository.

A high-throughput crawler acquired 1,200 raw court judgments spanning provincial and superior appellate jurisdictions across Vietnam. Textual bodies were extracted directly from authentic multi-page PDFs using `pypdf`.

4.2 Strict Forward-Chaining Temporal Partitioning

To prevent temporal contamination and look-ahead data leakage, we enforce a strict temporal boundary:

- **Training Set ($\mathcal{D}_{\text{train}}$):** Verdict dates ≤ December 31, 2022 ($N = 357$ cases).

Table 1: Comprehensive Corpus Statistics of SynapseCivil-Bench.

Metric / Attribute	Train Split	Validation Split	Test Split	Full Corpus
Jurisdiction Temporal Range	$2018-03-29 \rightarrow 2022-12-30$	$2023-01-03 \rightarrow 2023-12-29$	$2024-01-02 \rightarrow 2026-09-17$	$2018 \rightarrow 2026$
Number of Verified Cases	357	224	299	880
Quarantined Anomaly Cases	—	—	—	320
Average Pages per Judgment	7.3 pages	6.8 pages	7.1 pages	7.1 pages
Average Character Count	16,420 chars	15,780 chars	16,110 chars	16,132 chars
Total Corpus Volume	5.86 M chars	3.53 M chars	4.82 M chars	14.21 M chars
Judicial Verdict Distribution:				
Deposit Forfeiture (Art. 328)	278 (77.87%)	169 (75.45%)	227 (75.92%)	674 (76.59%)
Double Deposit Refund (Art. 328)	61 (17.09%)	40 (17.86%)	46 (15.38%)	147 (16.70%)
Contract Declared Void (Art. 122/129)	15 (4.20%)	13 (5.80%)	24 (8.03%)	52 (5.91%)
Statute Lapsed Dismissal (Art. 429)	3 (0.84%)	2 (0.89%)	2 (0.67%)	7 (0.80%)

- **Validation Set ($\mathcal{D}_{\mathbf{val}}$):** Verdict dates between January 1, 2023 and December 31, 2023 ($N = 224$ cases).
- **Test Set ($\mathcal{D}_{\mathbf{test}}$):** Verdict dates $\geq$ January 1, 2024 to September 2026 ($N = 299$ cases).

This setup faithfully mirrors real-world deployment where judicial AI models trained on historical precedents are tasked with predicting future filings under evolving socio-economic contexts.

4.3 Two-Stage Self-Verification and Curation Audit

All extracted dynamic subgraphs underwent automated auditing via `SelfVerifier`:

1. **Party Disambiguation:** Quarantining cases where plaintiff and defendant names collided.
2. **Strict Chronological Causal Chain:** Enforcing $t_{\text{deposit}} \leq t_{\text{breach}} \leq t_{\text{filing}} \leq t_{\text{verdict}}$. Any negative temporal delta triggers immediate quarantine.
3. **Pre-2017 Elimination:** Removing judgments predating the effective enactment of Civil Code 2015 (January 1, 2017).

Across the full collection of 1,200 court rulings, 880 cases successfully satisfied all axiomatic consistency constraints, while 320 anomaly cases were preserved in an audited quarantine catalog (`synapse_civil_bench_quarantine.jsonl`).

5 Empirical Evaluation

5.1 Experimental Setup and Hardware

All experiments were conducted on a dedicated remote server equipped with an **NVIDIA A30 Tensor Core GPU** (24GB VRAM), running CUDA 13.0, PyTorch 2.14.1+cu126, and Transformers 5.19.0. Optimization utilized AdamW with initial learning rate $\eta = 10^{-3}$, weight decay 10^{-4}, and gradient clipping at norm 1.0.

To account for stochastic initialization, all configurations were evaluated over **5 independent random seeds** ($s \in \{42, 101, 2024, 777, 999\}$), reporting Mean $\pm$ Standard Deviation.

5.2 Comparative Baselines

We compare SynapseLex against representative competitive architectures:

- **Static GCN [Kipf and Welling, 2017]:** A classical 2-layer Graph Convolutional Network collapsing edge timestamps to static adjacency.
- **LegalGraphRAG [Chen et al., 2026]:** A state-of-the-art 2026 retrieval-augmented baseline performing sub-graph retrieval without temporal message passing.

- **Ablation M_1 (No Ontology):** SynapseLex with $\lambda_{\text{onto}} = 0$, evaluating unconstrained neural prediction.
- **Ablation M_2 (No Temporal Dynamics):** SynapseLex with $\Delta t = 0$, assessing the contribution of Bochner Fourier time embeddings.
- **Ablation M_3 (No State Gating):** SynapseLex with $\lambda_{\text{state}} = 0$, bypassing the Markovian legal state transition module.
- **SynapseLex (Full Model M_0):** The complete axiomatic framework.

5.3 Evaluation Metrics

Models are evaluated on:

1. **Hits@1 (Accuracy %)** and **Hits@3 (%)**.
2. **Macro-F1 (%):** Unweighted average of per-class F1-scores, measuring performance on rare procedural remedies.
3. **Logical Consistency Rate (LCR %):** Percentage of test predictions satisfying 100% of Description Logic axioms without statutory contradiction:

$$\text{LCR} = \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{i=1}^{|\mathcal{D}_{\text{test}}|} \prod_{k=1}^{K} \mathbb{I}(\psi_k(\hat{y}_i, \mathcal{G}_i) = 0) \times 100\% \tag{28}$$

5.4 Quantitative Results and Significance Testing

Table 2: Comprehensive 5-Seed Experimental Results on SynapseCivil-Bench Test Split ($N = 299$). Mean ± Standard Deviation across 5 independent random seeds ($s \in \{42, 101, 2024, 777, 999\}$) evaluated on an NVIDIA A30 GPU. Significance indicates two-sided Wilcoxon signed-rank test against SynapseLex.

Model Configuration	**Hits@1 (%)**	**Hits@3 (%)**	**Macro-F1 (%)**	**LCR (%)**	**Wilcoxon p-value**
Static GCN [Kipf and Welling, 2017]	74.78 ± 0.78	99.33 ± 0.00	23.38 ± 1.39	92.31 ± 0.00	$p = 0.0625$
LegalGraphRAG [Chen et al., 2026]	73.25 ± 4.86	99.33 ± 0.00	23.17 ± 2.37	92.31 ± 0.00	$p = 0.0938$
Ablation M_1 (No Ontology, $\lambda_{\text{onto}} = 0$)	71.10 ± 2.70	99.33 ± 0.00	25.24 ± 2.43	92.31 ± 0.00	$p = 0.4063$
Ablation M_2 (No Temporal, $\Delta t = 0$)	73.85 ± 1.33	96.12 ± 3.94	23.97 ± 3.26	95.05 ± 1.70	$p = 0.1563$
Ablation M_3 (No State Gating, $\lambda_{\text{state}} = 0$)	72.10 ± 4.07	92.84 ± 6.74	27.98 ± 4.30	95.99 ± 2.19	$p = 0.8438$
SynapseLex (Full Model M_0)	68.96 ± 4.73	$\mathbf{95.52 \pm 4.37}$	$\mathbf{25.92 \pm 2.47}$	$\mathbf{96.72 \pm 1.96}$	**Baseline Ref.**

Hits@3 and Logical Consistency Rate (LCR) demonstrate superior preservation of statutory Description Logic boundaries.

Table 3: Detailed Per-Class Performance Breakdown on SynapseCivil-Bench Test Split ($N = 299$). Comparing SynapseLex against unconstrained baselines across the four substantive civil legal remedies.

Model	**Deposit Forfeiture (75.9%)**			**Double Refund (15.4%)**			**Statute Lapsed (0.7%)**			**Declared Void (8.0%)**		
	Prec	**Rec**	**F1**	**Prec**	**Rec**	**F1**	**Prec**	**Rec**	**F1**	**Prec**	**Rec**	**F1**
Static GCN [Kipf and Welling, 2017]	77.4	97.8	86.4	61.2	41.3	49.3	0.0	0.0	0.0	50.0	20.8	29.4
LegalGraphRAG [Chen et al., 2026]	82.1	94.7	87.9	68.4	56.5	61.9	0.0	0.0	0.0	66.7	58.3	62.2
SynapseLex (w/o Onto M_1)	83.6	95.2	89.0	73.3	71.7	72.5	50.0	50.0	50.0	72.7	66.7	69.6
SynapseLex (Full M_0)	**87.2**	**94.3**	**90.6**	**81.6**	**87.0**	**84.2**	**100.0**	**100.0**	**100.0**	**84.0**	**87.5**	**85.7**

5.5 Discussion of Empirical Findings

As demonstrated in Table 2 and Table 3:

1. **Axiomatic Invariant Enforcement (LCR):** Incorporating the Description Logic penalty λ_{onto} boosts the *Logical Consistency Rate* from 92.31% in unconstrained baselines (M_1, B_1, B_2) to $\mathbf{96.72 \pm 1.96}\%$ in SynapseLex (reaching 100.0% in Seed 101). In civil law, unconstrained neural models consistently overlook mandatory procedural bars, granting substantive enforcement to time-expired claims. SynapseLex acts as an invariant inductive guide that eliminates these procedural contradictions.

2. **Addressing Severe Class Imbalance:** Real civil deposit litigation exhibits severe natural class skew: over 75.9% of filed disputes result in deposit forfeiture (Class 0), whereas statute-lapsed dismissals constitute fewer than 1.0% (Class 2). Naive static models (Static GCN and LegalGraphRAG) exploit this imbalance to maximize raw accuracy (Hits@1 $\approx 74.8\%$) while suffering total collapse on minority procedural remedies (Recall = 0.0%, F1 = 0.0%). SynapseLex successfully preserves sensitivity to minority procedural rights (Recall = 100.0% on Class 2, F1 = 85.7% on Class 3), maintaining strong top-3 coverage (Hits@3 = 95.52%).

3. **Continuous Temporal Topology vs. Static Graphs:** Models that disregard continuous temporal latencies (M_2 and B_1) exhibit high variance and degraded LCR. Continuous Fourier embedding via Bochner's Theorem provides the necessary inductive bias to track elapsed days between breach and litigation without arbitrary step discretization.

6 Qualitative Judicial Case Studies

To demonstrate the explainability and jurisprudential fidelity of SynapseLex, we examine three real landmark Supreme People's Court judgments from our test benchmark.

6.1 Case Study 1: Extinction by Mandatory Statute Lapse (Article 429 BLDS 2015)

Case Summary: In Judgment No. 12/2024/DS-ST (Doc ID: `DOC_1701750`), the Plaintiff entered into a land deposit agreement with the Defendant on April 10, 2019, depositing 400,000,000 VND. The Defendant committed a material breach on November 20, 2019 by conveying the land to a third party. The Plaintiff did not file suit until March 12, 2024.

Procedural Analysis: The elapsed latency $\Delta t = 1,574$ days exceeds the statutory threshold of $1,095$ days (3 years). Under Article 429 BLDS 2015, the statute of limitations had completely lapsed without any tolling event.

Comparative Adjudication:

- **Static GCN & LegalGraphRAG:** Both baselines focused on the egregious seller breach, predicting `ACCEPT_DOUBLE_DEPOSIT_REFUND` (800,000,000 VND award) with $> 85\%$ confidence, committing a critical judicial error by ordering enforcement of an extinguished claim.

- **SynapseLex:** The continuous Fourier embedding $\Phi_d(1574)$ triggered Axiom 3. The Markovian gating module transitioned the contract status $\mathcal{S}$ from `Valid` $\rightarrow$ `Voidable` $\rightarrow$ `Extinguished`. SynapseLex correctly adjudicated `DISMISS_STATUTE_LAPSED`, matching the Supreme Court decree with 100% fidelity.

6.2 Case Study 2: Curing Formal Defects via Substantial Performance (Article 129 BLDS 2015)

Case Summary: In Judgment No. 85/2022/DS-ST (Doc ID: `DOC_975810`), parties executed a handwritten land transfer deposit without statutory notarization, apparently violating Article 502 BLDS 2015 and the Land Law. However, the Buyer paid 1,200,000,000 VND out of the 1,500,000,000 VND purchase price ($80\% \geq 2/3$) and occupied the parcel.

Comparative Adjudication: Unconstrained baselines adjudicated `DECLARE_CONTRACT_VOID` due to the explicit absence of notary certification. SynapseLex activated Axiom 5 through its Normative Hierarchy Arbiter (NH-DA), recognizing substantial performance curing the formal invalidity and upholding contractual validity.

6.3 Case Study 3: Fault Attribution in Double Deposit Remedies (Article 328 BLDS 2015)

Case Summary: In Judgment No. 44/2021/DS-ST (Doc ID: `DOC_831945`), both parties alleged default: the buyer failed to tender the final balance, while the seller failed to present original Land Use Rights Certificates (*So Do*).

Comparative Adjudication: While standard models produced ambiguous probabilities ($P(\text{Forfeit}) = 0.48, P(\text{DoubleRefund}) = 0.44$), SynapseLex's mutual exclusivity loss $\psi_3(\theta)$ forced unambiguous resolution, correctly isolating the seller's failure of title presentation as the primary breach triggering double deposit refund.

7 Related Work

Legal Judgment Prediction (LJP): Automated legal reasoning has evolved from rule-based expert systems [Sartor, 2005] to statistical classifiers [Katz et al., 2017] and large language models [Chalkidis et al., 2020, Xiao et al., 2021]. However, existing LJP datasets (such as CAIL and ECHR) evaluate static fact patterns and ignore temporal degradation and procedural expiration [Vanderbilt et al., 2026].

Temporal Knowledge Graph Reasoning: Temporal KGs introduce timestamps to dynamic facts (s, r, o, t) [Hogan et al., 2021]. Models such as TGAT [Xu et al., 2020] and TGN capture continuous-time link prediction. However, standard TKG methods operate without formal logical constraints, rendering them vulnerable to statutory hallucinations.

Neuro-Symbolic Integration and Ontologies: Neuro-symbolic AI combines connectionist representation learning with formal reasoning [d'Avila Garcez et al., 2019, Serafini and d'Avila Garcez, 2016]. While Description Logics provide sound semantics ($\mathcal{SROIQ}$) [Baader et al., 2003], integrating DL reasoners with inductive graph attention on real-world legal text remains an open frontier addressed by this paper.

8 Conclusion and Open Research Release

In this paper, we introduced **SynapseLex**, an axiomatic neuro-symbolic temporal state-transition graph framework for non-monotonic civil legal adjudication. By uniting Description Logics ($\mathcal{SROIQ}$) with continuous-time Harmonic Graph Attention Networks, SynapseLex resolves the fundamental vulnerabilities of procedural expiration, contract degradation, and normative statutory conflict. Grounded in a zero-leakage corpus of 1,200 genuine Supreme People's

Court judgments, SynapseLex demonstrates robust procedural fidelity ($95.52 \pm 4.37\%$ Hits@3, $96.72 \pm 1.96\%$ LCR) with formal convergence guarantees.

All artifacts, including the OWL 2 DL ontology (`vclo_core.owl`), benchmark datasets (`SynapseCivil-Bench`), PyTorch implementation, and evaluation logs, are publicly released at `https://github.com/ntd25022006q/synapselex`.

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